

# Health Data Sharing Platform

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A decentralized application (dApp) for securely sharing health data between patients and healthcare providers, built on the Ethereum Sepolia testnet.

## Tech Stack

- **Smart Contracts:** Solidity ^0.8.20
- **Development Framework:** Hardhat
- **Frontend:** React.js (Vite) + TailwindCSS
- **Web3 Library:** ethers.js v5
- **Off-chain Storage:** IPFS (via Pinata)
- **Network:** Ethereum Sepolia Testnet

## Project Structure

```
health-data-platform/  
├── contracts/                # Solidity smart contracts  
│   ├── PatientRegistry.sol  
│   ├── HealthRecordStorage.sol  
│   └── ConsentManager.sol  
├── scripts/                 # Deployment scripts  
│   └── deploy.js  
├── test/                   # Contract test suites  
├── frontend/               # React + Vite frontend  
│   └── src/  
│       ├── config/         # Contract addresses & ABIs  
│       ├── hooks/          # Custom React hooks  
│       ├── components/     # UI components  
│       └── utils/           # IPFS helpers, etc.  
├── hardhat.config.js  
└── package.json
```

## Smart Contracts

| Contract            | Purpose                                                    |
|---------------------|------------------------------------------------------------|
| PatientRegistry     | Registers patients and doctors and manages their identity. |
| HealthRecordStorage | Stores IPFS hashes of health records on-chain.             |
| ConsentManager      | Manages access permissions between patients and doctors.   |

## Getting Started

### 1. Install dependencies

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```
# Root (Hardhat)
npm install

# Frontend
cd frontend
npm install
```

## 2. Configure environment

Copy `.env.example` to `.env` and fill in:

- `SEPOLIA_RPC_URL` (from Alchemy or Infura)
- `PRIVATE_KEY` (deployer wallet — never commit a real key)
- `ETHERSCAN_API_KEY`
- Pinata credentials for IPFS

## 3. Compile contracts

```
npm run compile
```

## 4. Run tests

```
npm test
```

## 5. Deploy

```
# Local Hardhat node
npm run node           # in one terminal
npm run deploy:local   # in another

# Sepolia
npm run deploy:sepolia
```

Copy the deployed addresses into `frontend/.env` (or root `.env` with the `VITE_*` keys).

## 6. Run the frontend

```
cd frontend
npm run dev
```

# License

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